Xiaocan Li

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Research Interests

Particle acceleration and plasma energization processes associated with magnetic reconnection, collisionless shocks, and plasma turbulence. Solar flares. Coronal Heating. Solar Energetic Particles. Interstellar medium. Cosmic rays.

Education

PhD in Space Science, The University of Alabama in Huntsville, 2011 - 2016
MS in Space Physics, Peking University, Beijing, China, 2008 - 2011
BS in Surveying and Mapping Engineering, Tongji University, Shanghai, China, 2004 - 2008

Research Experiences

Dartmouth College — Hanover, NH
Research Associate B — *Feb. 2020 – present*
I study particle energization processes in solar physics and astrophysics. I perform both MHD and particle-in-cell simulations, develop theories of particle acceleration processes, and compare the results with emission observations.

Los Alamos National Laboratory — Los Alamos, NM
Postdoctoral Research Associate — *Feb. 2017 – Feb. 2020*
I have been studying particle acceleration during magnetic reconnection in solar flares. Specifically, I study the major particle acceleration mechanisms using kinetic particle-in-cell simulations, develop particle acceleration models, solve the models coupled with MHD simulations, and compare the results with solar flare observations.

The University of Alabama in Huntsville — Huntsville, AL
Postdoctoral Research Assistant — *Sept. 2016 – Jan. 2017*
I studied the interaction of plasma with electric tether using kinetic particle-in-cell simulations.

Los Alamos National Laboratory — Los Alamos, NM
Visiting Student — *May 2014 – Aug. 2016*
I studied particle energization during magnetic reconnection using the particle-in-cell code VPIC. Specifically, I studied the roles of particle drift motions, flow compression and shear, and pressure anisotropy in particle energization.

The University of Alabama in Huntsville — Huntsville, AL
Graduate Research Assistant — *May 2012 – Aug. 2016*
My dissertation is on numerical simulations of particle acceleration during solar flares. I developed a test-particle code to track charged particles in chaotic and turbulent magnetic fields, and I used the kinetic particle-in-cell code VPIC to study particle acceleration during magnetic reconnection.

Los Alamos National Laboratory/Center for Nonlinear Studies — Los Alamos, NM
Graduate Research Assistant — *Jun. 2013 – Aug. 2013*
I studied the magnetic field amplification in supernova remnant shocks using MHD simulations, and particle acceleration in the shock downstream by solving the Parker transport equation coupled with the MHD fields.

Peking University — Beijing, China
Graduate Research Assistant — *Sept. 2009 – Jun. 2011*
I studied the electron transport using the Monte Carlo code Geant4.

Grants and Awards

- Postdoctoral Distinguished Performance Award Honorable Mention, Los Alamos National Lab, 2019
- Institutional Computing Project 2019, Los Alamos National Lab, 10 million core-hours/year, two years

- Institutional Computing Project 2018, Los Alamos National Lab, 8.8 million core-hours/year, two years
- Co-I of NERSC's High-Impact Science at Scale program 2018, 60 millions core-hours
- Co-I of Fermi Guest Investigator Program Cycle 11, 2018
- NASA Earth and Space Science Fellowship (NESSF), NASA, $30,000/year, 2013-2016
- The Argonne Training Program on Extreme-Scale Computing (ATPESC), Participant, 2016
- SPD Studentship Award for Solar Physics Division Meeting, AAS, 2015
- DPP Student Travel Grant Award for 56th Annual Meeting of APS Division of Plasma Physics, APS, 2014

Service and Professional Membership

- Referee for ApJ, ApJL, ApJS, MNRAS, JGR, GRL, PoP, Astrophysics and Space Sciences, Solar Physics, Plasma Science and Technology, Galaxies, and Sensors
- Convener for 2019 AGU Fall Meeting (sessions: SH12B and SH13D on "Plasma Energization, Particle Acceleration, and High-Energy Emission in Solar Flares")
- Organizer of the 2019 workshop on "Magnetic Reconnection and Particle Acceleration in Solar Flares"
- Session Chair for 2018 AGU Fall Meeting (sessions: SH11D and SH14A on "Plasma Energization, Particle Acceleration, and High-Energy Emission in Solar Flares")
- Judge for Outstanding Student Paper Award of the AGU Fall Meeting 2017–2019
- Reviewer for NASA grant programs
- Member of AGU and APS

Computer Skills

- Seven-year experiences of using supercomputers at LANL, NERSC, and TACC.
- Languages: C, C++, Python, and Fortran. Parallelization: MPI, OpenMP, and CUDA. Building tools: Make and CMake. Debugging tools: GDB, Valgrind, and Arm DDT. Version control: Git. I/O: HDF5.
- Data analysis and visualization tools: Python libraries, IDL, Matlab, FFTW3, and ParaView.
- Familiar codes: Athena/Athena++ MHD code, LA-COMPASS MHD code, VPIC particle-in-cell code, test-particle code (`https://git.io/fxQYy`), stochastic code to solve the Parker transport equation (`https://git.io/fxQY1`).

Teaching Experiences

Department of Physics, The University of Alabama in Huntsville — Huntsville, AL
Graduate Teaching Assistant — *Spring 2012*
Grading for PH421 (Thermal & Statistical Physics)

Department of Physics, The University of Alabama in Huntsville — Huntsville, AL
Graduate Teaching Assistant — *Fall 2011*
Grading for PH113 (General Physics with Calculus III) and teaching PH114 (General Physics Labs)

Journal Articles

23. Fan Guo, **Xiaocan Li**, William Daughton, Hui Li, Patrick Kilian, Yi-Hsin Liu, Qile Zhang, and Haocheng Zhang. "Magnetic Energy Release, Plasma Dynamics and Particle Acceleration during Relativistic Turbulent Magnetic Reconnection." submitted to The Astrophysical Journal, arXiv preprint arXiv:2008.02743 (2020).
22. Fan Guo, Yi-Hsin Liu, **Xiaocan Li**, Hui Li, William Daughton, and Patrick Kilian. "Recent progress on particle acceleration and reconnection physics during magnetic reconnection in the magnetically-dominated relativistic regime." Physics of Plasmas 27, no. 8 (2020): 080501.
21. Patrick Kilian, **Xiaocan Li**, Fan Guo, and Hui Li. "Exploring the acceleration mechanisms for particle injection and power-law formation during trans-relativistic magnetic reconnection." submitted to The Astrophysical Journal, arXiv preprint arXiv:2001.02732 (2020).

20. Senbei Du, Gary P. Zank, Xiaocan Li, and Fan Guo. "Energy dissipation and entropy in collisionless plasma." Physical Review E 101, no. 3 (2020): 033208.
19. Yi-Hsin Liu, Shan-Chang Lin, Michael Hesse, Fan Guo, **Xiaocan Li**, Haocheng Zhang, and Sarah Peery. "The Critical Role of Collisionless Plasma Energization on the Structure of Relativistic Magnetic Reconnection." The Astrophysical Journal Letters, 892 L13
18. Fu, Xiangrong, Fan Guo, Hui Li, and **Xiaocan Li**. "Heating of Heavy Ions in Low-beta Compressible Turbulence." The Astrophysical Journal, 890(2), 161.
17. **Xiaocan Li**, Fan Guo, Hui Li, Adam Stanier, and Patrick Kilian. "Formation of Power-law Electron Energy Spectra in Three-dimensional Low-β Magnetic Reconnection." The Astrophysical Journal 884, no. 2 (2019): 118.
16. Chen, G., H. S. Fu, Y. Zhang, **Xiaocan Li**, Y. S. Ge, A. M. Du, C. M. Liu, and Y. Xu. "Energetic Electron Acceleration in Unconfined Reconnection Jets." The Astrophysical Journal Letters 881, no. 1 (2019): L8.
15. Adam Stanier, William Daughton, Ari Le, **Xiaocan Li**, and Robert Bird. "Influence of 3D plasmoid dynamics on the transition from collisional to kinetic reconnection." Physics of Plasmas 26, 072121 (2019)
14. Fan Guo, **Xiaocan Li**, William Daughton, Hui Li, Yi-Hsin Liu, Wangcheng Yan, Dylan Ma, and Patrick Kilian. "Determining the Dominant Acceleration Mechanism during Relativistic Magnetic Reconnection in Large-scale Systems." The Astrophysical Journal Letters 879, no. 2 (2019): L23.
13. **Xiaocan Li**, Fan Guo, Hui Li. "Particle acceleration in kinetic simulations of non-relativistic magnetic reconnection with different ion-electron mass ratio" The Astrophysical Journal, 879:5 (12pp), 2019 July 1.
12. Senbei Du, Fan Guo, Gary P. Zank, **Xiaocan Li**, and Adam Stanier. "Plasma Energization in Colliding Magnetic Flux Ropes." The Astrophysical Journal 867, no. 1 (2018): 16.
11. **Xiaocan Li**, Fan Guo, Hui Li, Shengtai Li. "Large-scale Compression Acceleration during Magnetic Reconnection in a Low-β Plasma." The Astrophysical Journal 866, no. 1 (2018): 4.
10. Haocheng Zhang, **Xiaocan Li**, Fan Guo, and Dimitrios Giannios. "Large-Amplitude Blazar Polarization Angle Swing as a Signature of Magnetic Reconnection." The Astrophysical Journal Letters 862, no. 2 (2018): L25.
9. **Xiaocan Li**, Fan Guo, Hui Li, Joachim Birn. "The roles of fluid compression and shear in electron energization during magnetic reconnection." The Astrophysical Journal, 855:80 (2018)
8. Xiangrong Fu, Hui Li, Fan Guo, **Xiaocan Li**, and Vadim Roytershteyn. "Parametric Decay Instability and Dissipation of Low-frequency Alfvén Waves in Low-beta Turbulent Plasmas." The Astrophysical Journal 855, no. 2 (2018): 139.
7. **Xiaocan Li**, Fan Guo, Hui Li, and Gang Li. "Particle Acceleration during Magnetic Reconnection in a Low-beta Plasma." The Astrophysical Journal 843.1 (2017): 21.
6. Fan Guo, Hui Li, William Daughton, **Xiaocan Li**, and Yi-Hsin Liu. "Particle acceleration during magnetic reconnection in a low-beta pair plasma." Physics of Plasmas 23, no. 5 (2016): 055708.
5. Fan Guo, **Xiaocan Li**, Hui Li, William Daughton, Bing Zhang, Nicole Lloyd-Ronning, Yi-Hsin Liu, Haocheng Zhang, and Wei Deng. "Efficient production of high-energy nonthermal particles during magnetic reconnection in a magnetically dominated ion–electron plasma." The Astrophysical Journal Letters 818, no. 1 (2016): L9.
4. **Xiaocan Li**, Fan Guo, Hui Li, and Gang Li. "Nonthermally Dominated Electron Acceleration during Magnetic Reconnection in a Low-beta Plasma." The Astrophysical Journal Letters, 811:L24 (5pp), 2015 October 1
3. **Xiaocan Li**, Brahmananda Dasgupta, and Gang Li. "Energization of charged particle in a time-dependent chaotic magnetic field with an implication of the production of seed particles in solar energetic particle events." Advances in Space Research 53, no. 8 (2014): 1153-1161.
2. Lorin Arnold, Gang Li, **Xiaocan Li**, and Yihua Yan. "Observation of flux-tube crossings in the solar wind." The Astrophysical Journal 766, no. 1 (2013): 2.
1. **Xiaocan Li**, Hongfei Chen, Yongqiang Hao, Hong Zou, and Weihong Shi. "Investigation of electrons inside the satellite by the Geant4 simulation." Science China Technological Sciences 54, no. 9 (2011): 2271-2275.

Conference Proceedings

4. Bin Dong, Patrick Kilian, **Xiaocan Li**, Fan Guo, Suren Byna, and Kesheng Wu. 2019. "Terabyte-scale Particle Data Analysis: An ArrayUDF Case Study". In Proceedings of the 31st International Conference on Scientific and Statistical Database Management (SSDBM '19). ACM, New York, NY, USA, 202-205.
3. Senbei Du, Gary P. Zank, Fan Guo, **Xiaocan Li**, and Adam Stanier. "Particle Acceleration in Interacting Magnetic Flux Ropes." In Journal of Physics: Conference Series, vol. 1100, no. 1, p. 012009. IOP Publishing, 2018.
2. Kirit Makwana, Hui Li, Fan Guo, and **Xiaocan Li**. "Dissipation and particle energization in moderate to low beta turbulent plasma via PIC simulations." In Journal of Physics: Conference Series, vol. 837, no. 1, p. 012004. IOP Publishing, 2017.
1. Brahmananda Dasgupta, Gang Li, **Xiaocan Li**, Abhay Ram, Qiang Hu, Gang Li, Gary P. Zank, Xianzhi Ao, Olga Verkhoglyadova, and James H. Adams. "Particle transport and acceleration in a chaotic magnetic field: Implications for seed population to solar flare and CME." In AIP Conference Proceedings-American Institute of Physics, vol. 1500, no. 1, p. 56. 2012.

Invited Presentations

14. Large-scale particle acceleration during magnetic reconnection in solar flares. *Solar Physics Webinar of Global Reach—SolFER Colloquium*, Aug. 21, 2020, Webex online
13. The formation of power-law energy spectrum in low-β magnetic reconnection. *Magnetic Reconnection and Particle Acceleration in Solar Flares*, Mar. 18-20, 2019, Los Alamos, NM
12. The formation of power-law energy spectrum in 3D low-β magnetic reconnection. *The 18th Annual International Astrophysics Conference*, Feb. 18-22, 2019, Pasadena, CA
11. Particle Acceleration during Magnetic Reconnection in Solar Flares. *Dartmouth College*, Oct. 23, 2018, Hanover, NH.
10. Particle acceleration mechanisms during magnetic reconnection. *National Space Science Center*, May 31, 2018, Beijing, China.
9. Particle acceleration mechanisms during magnetic reconnection. *Beihang University*, May 30, 2018, Beijing, China.
8. Particle acceleration mechanisms during magnetic reconnection. *Shandong University Weihai Campus*, May 23, 2018, Weihai, Shangdong, China.
7. Particle acceleration mechanisms during magnetic reconnection. *Purple Mountain Observatory*, May 16, 2018, Nanjing, Jiangsu, China.
6. The Roles of Fluid Compression and Shear in Particle Energization during Magnetic Reconnection. *Particle Transport and Energization in Turbulent Plasmas*, April 24-27, 2018, Zhuhai, Guangdong, China.
5. The Roles of Fluid Compression and Shear in Particle Energization during Magnetic Reconnection. *The 16th Annual International Astrophysics Conference*, March 5-9, 2018, Santa Fe, NM.
4. The Role of Fluid Compression in Particle Energization during Magnetic Reconnection. *2017 AGU Fall Meeting*, New Orleans, LA.
3. The Role of Fluid Compression in Particle Energization during Magnetic Reconnection. *The 16th Annual International Astrophysics Conference*, March 6-10, 2017, Santa Fe, NM
2. Nonthermally Dominated Electron Acceleration during Magnetic Reconnection in a Low-beta Plasma. *Plasma Energization: Exchanges between Fluid and Kinetic Scales*, May 4-6, 2015, Los Alamos, NM.
1. Particle acceleration in reconnection: a view from electric current, *Meso-scale Plasma Dynamics and Energetic Particles: Applications to Laboratory, Space, and Astrophysical Plasmas*, June 30 – July 1, 2014, Los Alamos, NM.

Other Presentations

25. Power-law formation of nonthermal electrons in low-β reconnection. *MMS telecom*, July 28, 2020, online Webex meeting (Talk)
24. Power-law formation in 3D low-β magnetic reconnection. *DRIVE Science Center Collaboration on Solar Flare Energy Release (SolFER), Group 3*, July 9, 2020, online Zoom meeting (Talk)

23. Large-scale particle acceleration by magnetic reconnection during solar flares, *2019 AGU Fall Meeting*, San Francisco, CA (Poster)
22. The formation of power-law energy spectrum in 3D low-β magnetic reconnection, *KITP Conference: Connecting Micro and Macro Scales: Acceleration, Reconnection, and Dissipation in Astrophysical Plasmas*, September 9th - 12th, 2019, Santa Barbara, CA (Poster)
21. The formation of power-law energy spectrum in 3D low-β magnetic reconnection *SHINE Conference 2019*, August 5th - 9th, 2019, Boulder, CO (Poster)
20. Large-scale particle acceleration during solar flare reconnection. *18th RHESSI Workshop*, May 28-June 1, 2019, Minneapolis, MN (Talk)
19. The formation of power-law particle energy spectrum in 3D low-beta magnetic reconnection. *2018 AGU Fall Meeting*, Washington, D.C. (Poster)
18. Particle acceleration in compressible reconnection layer. *60th Annual Meeting of the APS Division of Plasma Physics*, November 5–9, 2018; Portland, Oregon. (Talk)
17. Large-scale particle acceleration by magnetic reconnection during solar flares. *2017 AGU Fall Meeting*, New Orleans, LA. (Poster)
16. The Role of Compression in Particle Energization during Magnetic Reconnection. *2017 JSI Workshop, Cosmic Accelerators: Understanding Nature's High-energy Particles and Radiation*, Annapolis, MD. (Poster)
15. Nonthermal particle acceleration in 2D and 3D nonrelativistic magnetic reconnection. *59th Annual Meeting of the APS Division of Plasma Physics*, October 23-27, 2017, Milwaukee, WI. (Talk)
14. The role of fluid compression in particle energization at the reconnection site during solar flares. *RHESSI 16th Workshop*, June 19-24, 2017, Boulder, CO. (Talk)
13. The Role of Fluid Compression in Particle Energization during Magnetic Reconnection. *2016 AGU Fall Meeting*, San Francisco, CA. (Poster)
12. Nonthermally Dominated Electron Acceleration during Magnetic Reconnection in a Low-beta Plasma. *SHINE Conference 2016*, July 11-15, 2016, Santa Fe, NM. (Poster)
11. Nonthermally Dominated Particle Acceleration during Magnetic Reconnection in a Low-beta Plasma. *2015 AGU Fall Meeting*, San Francisco, CA. (Poster)
10. The Role of Fluid Compression in Particle Acceleration during Magnetic Reconnection. *57th Annual Meeting of the APS Division of Plasma Physics*, November 16-20, 2015, Savannah, GA. (Poster)
9. Nonthermally Dominated Particle Acceleration during Magnetic Reconnection in a Low-beta Plasma. *14th RHESSI Workshop*, July 12 – 15, 2015, Newark, NJ. (Poster & Talk)
8. Nonthermally Dominated Electron Acceleration during Magnetic Reconnection in a Low-beta Plasma. *SHINE Conference 2015*, July 6 – 10, 2015, Stowe, VT. (Poster)
7. Particle Energization during the Relaxation of a Force-free Magnetic Field. *Triennial Earth Sun-Summit meeting*, April 26 – 30, 2015, Indianapolis, IN. (Poster)
6. Nonthermally Dominated Electron Acceleration during Magnetic Reconnection in a Low-beta Plasma. *Triennial Earth Sun-Summit meeting*, April 26 – 30, 2015, Indianapolis, IN. (Poster)
5. Particle energization in a force-free/non-force-free magnetic field. Abstract ID: SH23B-4158, *2014 AGU Fall Meeting*, San Francisco, CA. (Poster)
4. On the energy conversion and particle acceleration during magnetic reconnection. Abstract ID: SM13A-4146, *2014 AGU Fall Meeting*, San Francisco, CA. (Poster)
3. On the energy conversion and particle acceleration during magnetic reconnection, Abstract ID: PP8.00137, *56th Annual Meeting of the APS Division of Plasma Physics*, October 27 – 31, 2014, New Orleans, LA. (Poster)
2. Seed population generation by time-dependent chaotic magnetic field in solar corona, Abstract SH21A-2075, *2013 AGU Fall Meeting*, San Francisco, CA. (Poster)
1. Particle acceleration and magnetic field amplification in supernova remnants shocks, *2013 LANL Student Symposium*, Los Alamos, NM. (Poster)